DUALITY EXACT SEQUENCES FOR A REDUCED CLOSED SUBSCHEME OF A GORENSTEIN CURVE AND ITS SCHEME-THEORETIC COMPLEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the union sequence, the four kernel descriptions with the diagram chase, the six structure-sheaf exact sequences, the reduction by symmetry, and the closing annihilator identification by evaluation at 1 are all retained verbatim.

Editorial note. The label lemma-closed-subscheme-reduced-gorenstein names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Scheme-theoretic unions and the union sequence. Morphisms, Lemma morphisms-lemma-scheme-theoretic-union. Supplies 0 -> O_X -> O_Y (+) O_Z -> O_{Y cap Z} -> 0 for X = Y union Z and the kernel descriptions of the ideal sheaves used in the diagram chase.
- Duality for 1-dimensional CM closed subschemes. Same chapter, Lemma lemma-duality-dim-1 for the dualizing modules and Lemma lemma-closed-immersion-dim-1-CM; Duality for Schemes, Lemma duality-lemma-gorenstein. Makes omega_X invertible on the Gorenstein X, identifies i_*omega_Y as SheafHom(i_*O_Y, omega_X), and trivializes omega_X over the 0-dimensional intersection.

Source reference. Original lemma 0BFQ: sources/stacks-chow/tags.txt, source lines 14393–14393. Prerequisite pointer invoked at proof line(s) 1081; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E33: sources/stacks-curves/pinned-original-source.tex, source lines 976–976. Prerequisite pointer invoked at proof line(s) 1093; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BS2: sources/stacks-curves/pinned-original-source.tex, source lines 796–796. Prerequisite pointer invoked at proof line(s) 1048; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0C4J: sources/stacks-chow/tags.txt, source lines 15229–15229. Prerequisite pointer invoked at proof line(s) 1055; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a proper scheme over a field k which is Gorenstein, reduced, and equidimensional of dimension 1. Let $i : Y \to X$ be a reduced closed subscheme equidimensional of dimension 1. Let $j : Z \to X$ be the scheme theoretic closure of $X \setminus Y$. Then*

(1) *Y and Z are Cohen-Macaulay,*
(2) *if $\mathcal{I} \subset \mathcal{O}_X$, resp. $\mathcal{J} \subset \mathcal{O}_X$ is the ideal sheaf of Y, resp. Z in X, then*
$$\mathcal{I} = i_*\mathcal{I}' \quad \text{and} \quad \mathcal{J} = j_*\mathcal{J}'$$
where $\mathcal{I}' \subset \mathcal{O}_Z$, resp. $\mathcal{J}' \subset \mathcal{O}_Y$ is the ideal sheaf of $Y \cap Z$ in Z, resp. Y,
(3) *$\omega_Y = \mathcal{J}'(i^*\omega_X)$ and $i_*(\omega_Y) = \mathcal{J}\omega_X$,*
(4) *$\omega_Z = \mathcal{I}'(i^*\omega_X)$ and $i_*(\omega_Z) = \mathcal{I}\omega_X$,*
(5) *we have the following short exact sequences*
$$\begin{aligned}
0 \to \omega_X \to i_*i^*\omega_X \oplus j_*j^*\omega_X \to \mathcal{O}_{Y\cap Z} \to 0 \\
0 \to i_*\omega_Y \to \omega_X \to j_*j^*\omega_X \to 0 \\
0 \to j_*\omega_Z \to \omega_X \to i_*i^*\omega_X \to 0 \\
0 \to i_*\omega_Y \oplus j_*\omega_Z \to \omega_X \to \mathcal{O}_{Y\cap Z} \to 0 \\
0 \to \omega_Y \to i^*\omega_X \to \mathcal{O}_{Y\cap Z} \to 0 \\
0 \to \omega_Z \to j^*\omega_X \to \mathcal{O}_{Y\cap Z} \to 0
\end{aligned}$$

Here ω_X, ω_Y, ω_Z are as in Lemma 0BS2.

Proof. A reduced 1-dimensional Noetherian scheme is Cohen-Macaulay, so (1) is true. Since X is reduced, we see that $X = Y \cup Z$ scheme theoretically. With notation as in Morphisms, Lemma 0C4J and by the statement of that lemma we have a short exact sequence
$$0 \to \mathcal{O}_X \to \mathcal{O}_Y \oplus \mathcal{O}_Z \to \mathcal{O}_{Y\cap Z} \to 0$$
Since $\mathcal{J} = \mathrm{Ker}(\mathcal{O}_X \to \mathcal{O}_Z)$, $\mathcal{J}' = \mathrm{Ker}(\mathcal{O}_Y \to \mathcal{O}_{Y\cap Z})$, $\mathcal{I} = \mathrm{Ker}(\mathcal{O}_X \to \mathcal{O}_Y)$, and $\mathcal{I}' = \mathrm{Ker}(\mathcal{O}_Z \to \mathcal{O}_{Y\cap Z})$ a diagram chase implies (2). Observe that $\mathcal{I}+\mathcal{J}$ is the ideal sheaf of $Y \cap Z$ and that $\mathcal{I} \cap \mathcal{J} = 0$. Hence we have the following exact sequences
$$\begin{aligned}
0 \to \mathcal{O}_X \to \mathcal{O}_Y \oplus \mathcal{O}_Z \to \mathcal{O}_{Y\cap Z} \to 0 \\
0 \to \mathcal{J} \to \mathcal{O}_X \to \mathcal{O}_Z \to 0 \\
0 \to \mathcal{I} \to \mathcal{O}_X \to \mathcal{O}_Y \to 0 \\
0 \to \mathcal{J} \oplus \mathcal{I} \to \mathcal{O}_X \to \mathcal{O}_{Y\cap Z} \to 0 \\
0 \to \mathcal{J}' \to \mathcal{O}_Y \to \mathcal{O}_{Y\cap Z} \to 0 \\
0 \to \mathcal{I}' \to \mathcal{O}_Z \to \mathcal{O}_{Y\cap Z} \to 0
\end{aligned}$$
Since X is Gorenstein ω_X is an invertible $\mathcal{O}_X$-module (Duality for Schemes, Lemma 0BFQ). Since $Y \cap Z$ has dimension 0 we have $\omega_X|_{Y\cap Z} \cong \mathcal{O}_{Y\cap Z}$. Thus if we prove (3) and (4), then we obtain the short exact sequences of the lemma by tensoring the above short exact sequence with the invertible module ω_X. By symmetry it suffices to prove (3) and by (2) it suffices to prove $i_*(\omega_Y) = \mathcal{J}\omega_X$.

We have $i_*\omega_Y = \mathcal{H}om_{\mathcal{O}_X}(i_*\mathcal{O}_Y, \omega_X)$ by Lemma 0E33. Again using that ω_X is invertible we finally conclude that it suffices to show $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{O}_X/\mathcal{I}, \mathcal{O}_X)$ maps isomorphically to $\mathcal{J}$ by evaluation at 1. In other words, that $\mathcal{J}$ is the annihilator of $\mathcal{I}$. This follows from the above. □